Haiyan Qin 秦海岩



Agent infrastructure and inference systems

Ph.D. student at Beihang University · Core developer of ApxInf

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qhy991.github.io | GitHub | Google Scholar

I build agent runtimes, workflow orchestration, and GPU evaluation infrastructure for verifiable, recoverable optimization of inference systems.

Education and internship

Beihang University · Ph.D. in Integrated Circuit Engineering Sep 2023 - Jul 2028 expected

Beihang University · B.Eng. in Microelectronics Science and Engineering 2019 - 2023

Infinigence AI · Research intern Sep 2025 - Present

Selected achievements

- Helped Infinigence AI achieve first place on two NVIDIA SOL-ExecBench leaderboards.
- Contributed to GLM-5.3 inference optimization, reaching 630 tokens/s.

Open source and engineering

ApxInf Core developer

- Develop an open-source Rust and CUDA inference engine for embodied edge AI, targeting devices including Jetson Thor and Orin.

KernelOwl / KerSor / AKW Lead designer and developer

- Built a ReAct runtime across CLI, TUI, and SDK, with MCP extensions, task graphs, sessions, traces, and context management.
- Designed typed task handoffs and host-side verification for Codex and Claude Code, with rollback and verified-best version tracking.

GPU Infra / Agent GPU Broker Lead designer and developer

- Built immutable candidate submission, shared/exclusive GPU scheduling, and atomic multi-GPU allocation across compilation, correctness, sanitizers, and timing.
- Validated on A800/B200; used SIGKILL fault injection on A800 to check process, container, and GPU cleanup while retaining interruption records.

Cake IR / open-cake-ir Lead designer and developer

- Designed a Python DSL and typed schedule/program IR for tiling, memory, and synchronization, with Triton, CUDA, CuTe DSL, and Metal code generation.

EngineTailor Lead developer and first author

- Designed Manager/Worker/Supervisor collaboration and worktree isolation, separating candidate generation from acceptance and verification.
- Optimized graph boundaries for Qwen3-4B on B200 at concurrency 16; output throughput rose from 3078.6 to 4654.8 tokens/s, a 51.2% gain within the same engine.

Selected publications

EngineTailor DAI 2026 · First author · Accepted

Minimal Complete Inference Engines for Agent-Driven Optimization

ReasoningV ICAIS 2025 · First author

Efficient Verilog Code Generation with Adaptive Hybrid Reasoning Model

Technical skills

Agents and systems: ReAct, MCP, task graphs, sessions/traces, state recovery, GPU scheduling, independent evaluation, Docker, Linux

Engineering and inference: Python, C++/CUDA, PyTorch, Triton, SGLang, MLX/Metal, Git